Automatic Sea-Ice Classification of SAR Images Based on Spatial and Temporal Features Learning

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***Abstract*— Sea ice has a significant effect on climate change and ship navigation. Hence, it is crucial to draw sea-ice maps that reflect the geographical distribution of different types of sea ice. Many automatic sea-ice classification methods using synthetic aperture radar (SAR) images are based on the polarimetric characteristics or image texture features of sea ice. They either require professional knowledge to design the parameters and features or are sensitive to noise and condition changes. Moreover, ice changes over time are often ignored. In this article, we propose a new SAR sea-ice image classification method based on a combined learning of spatial and temporal features, derived from residual convolutional neural networks (ResNet) and long short-term memory (LSTM) networks. In this way, we achieve automatic and refined classification of sea-ice types. First, we construct a seven-type ice data set according to the Canadian Ice Service ice charts. We extract spatial feature vectors of a time series of sea-ice samples using a trained ResNet network. Then, using the feature vectors as inputs, the LSTM network further learns the variation of the set of sea-ice samples with time. Finally, the extracted high-level features are fed into a softmax classifier to output the most recent ice type. Taking both spatial features and time variation into consideration, our method can achieve a high classification accuracy of 95.7% for seven ice types. Our method can automatically produce more objective sea-ice interpretation maps, allowing detailed sea-ice distribution and improving the efficiency of sea-ice monitoring tasks.**

***Index Terms*— Ice charts, long short-term memory (LSTM), residual convolution network, sea-ice classification, synthetic aperture radar (SAR).**

I. Introduction

MONITORING of Antarctic and Arctic ice is of great significance to global climate change and atmospheric circulation research [1]. Sea-ice classification is crucial for sea-ice monitoring, and the result can be used to extract the extent and concentration of the ice.

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In recent decades, synthetic aperture radar (SAR) has become an indispensable tool for sea-ice monitoring because SAR satellites can gather multiview data day and night through all weather conditions. Different types of sea ice have variant dielectric properties and surface features, which form different backscattering characteristics and textures on the SAR data. Complex environmental (such as temperature, humidity, and wind speed) and radar wave characteristics (such as wavelength, polarization, and incident angle) are important factors affecting the response of sea ice to electromagnetic waves. Hence many researchers have classified sea-ice types based on the polarimetric characteristics of sea ice or extracted texture features in different wavebands, polarization modes, and seasons, using a statistical classifier such as Wishart [2] or some common machine learning algorithms such as neural network, support vector machine (SVM), and Markov random fields (MRFs) [3]–[8]. These sea-ice classification methods require professional knowledge to generate human-designed and low-level parameters and features and may also be sensitive to noise and condition changes. The robustness problem of the methods has also led to the fact that, although the accuracy of these studies is very high, people in the polar region rely more heavily on hand-drawn ice charts, which takes a lot of time and labor.

On the other hand, according to the Canadian Ice Service (CIS), the European Sentinel mission will lead to significantly increased volumes of SAR imagery, which demands new and fully automated image analysis methods [9]. Deep learning (DL) technology has attracted high attention in recent years, especially the performance of convolutional neural networks (CNNs), and their variations for image classification, are outstanding [10]–[12]. DL can automatically learn object features through a large number of sample data and has strong mobility and universality. Although DL models have been successfully used on land-cover and oceanic phenomena identification with SAR images [13]–[15] and ice concentration estimation [16]–[18], research on DL-based sea-ice classification of SAR images is limited [19]. Specifically, we address the following challenges.

1) On SAR images, different types of sea ice show little difference. As shown in Fig. 1, an example of SAR image with CIS's ice interpretation, we can hardly distinguish the sea-ice types without professional training. The variation of different sea-ice types on the SAR image is very small, especially between the ice types in closely related development stages, such as thin first-year (FY) ice and medium FY ice. Moreover, the complex sea-ice

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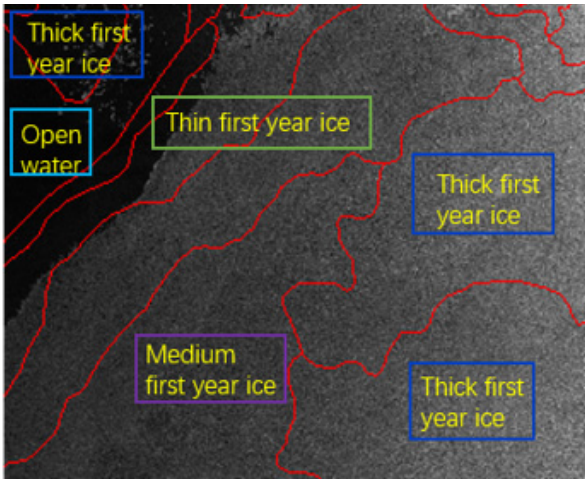


Fig. 1. Labeled sea-ice types on a Sentinel-1 A SAR image according to a CIS ice chart. The SAR image was acquired in Hudson Bay on May 20, 2017.

environment will also affect the sea-ice recognition of SAR images, such as surface roughness, moisture, and salinity [20].

2) A large data set is needed to train a DL model. At present, most of the sea-ice classification methods are supervised, and their performance highly depends on the quality of labels (also called ground truth) [21]–[23]. Most studies rely on ice charts that are produced by manual interpretation of various data sources, such as ships, air observations, remote sensing, and model outputs. However, the ice charts only provide regional information of ice types about their development stages. The National Ice Center in USA can provide digitalized data of ice charts, but just for four ice types.
3) The variability of ice signatures with seasonal conditions limits the use of automatic classification algorithms for ice charting and operational purposes [24]. How to capture time impact on sea ice should be considered. According to the development stage of sea ice, the growth and evolution of sea ice are periodic in a region. To specify this, we selected 799 weekly ice charts of CIS from five areas of Hudson Bay between January 2014 and January 2018. By recording the sea-ice type at the current time and the following week, we made statistics on the transfer possibilities from one type to the others. The results are shown in Table I. Through these statistics, the regularity of ice type transfer is clear: the type of sea ice in the same area is most likely to remain unchanged in the following week; if the status is changed, the ice will most likely transfer to the closed development stage (e.g., it will become thicker when freezing or thinner when melting). In general, the sea ice has obvious time-related transformation characteristics; however, this simple fact cannot be captured by sea-ice-type classification methods based on SAR images with single time resolution.

In view of the above challenges, in this article, we propose an end-to-end DL-based sea-ice classification method with SAR images, which can automatically identify seven sea-ice types. To classify the detailed sea-ice types at different periods, the method considers both spatial and temporal information of sea ice by combining a residual convolutional network (ResNet) [10], [25] and a long short-term memory (LSTM) [26] network. To train and test the DL framework, we constructed a seven-type sea-ice data set with dual-polarization Sentinel-1 SAR images, using the CIS ice charts as ground truth.

TABLE I

STATISTICS RESULT OF CIS ICE CHART (OW: OPEN WATER, NI: NEW ICE, GI: GRAY ICE, GWI: GRAY-WHITE ICE, THINFI: THIN FY ICE, MFI: MEDIUM FY ICE, AND THICKFI: THICK FY ICE)

	OW	NI	GI	GWI	ThinFI	MFI	ThickFI
OW	74.48	8.97	6.21	0.00	0.00	2.07	8.28
NI	5.00	20.00	55.00	20.00	0.00	0.00	0.00
GI	2.08	6.25	52.08	33.33	6.25	0.00	0.00
GWI	0.00	0.00	1.45	62.32	30.43	2.90	2.90
ThinFI	0.88	0.00	0.88	6.19	70.80	20.35	0.88
MFI	3.33	0.00	0.67	1.33	4.67	78.00	12.00
ThickFI	11.65	0.00	0.00	0.00	0.00	1.61	86.75

In our experiments, the proposed method could achieve 86.75% of classification accuracy when only a ResNet was used and 95.69% when combining ResNet with LSTM. Applying the classification framework, we can generate sea-ice distribution maps with a time series of SAR images, which are consistent with CIS ice charts, and provide more detailed (pixel-block level) sea-ice information than the ice charts. Compared with the human-involved interpretation of sea-ice classification, this method is automatically learning, easier to operate, and less subjective (i.e., it is not sensitive to human's lack of specialism).

The related work about sea-ice classification with SAR imagery is introduced in Section II. Study area and data set preparation are introduced in Section III. The methodology is presented in Section IV. In Section V, the comprehensive experiments and analysis are presented. Section VI draws the main conclusions.

II. RESEARCH BACKGROUND

A. Definition of Sea-Ice Types

There are no strict rules for sea-ice types. Depending on operational demands or research interests, sea-ice types have been classified from different perspectives, such as sea-ice development stage, sea-ice movement state, floe size, and sea-ice surface roughness. The development stage of sea ice is actually based on the ice thickness, which can divide the sea ice into 4–5 stages, roughly, or over ten stages, precisely. Table II lists the development stages of some sea-ice types with their thickness and codes, which are defined by CIS [27]. For example, FY ice, coded as "6," represents ice having thickness $\geq$30 cm, and it can be further subdivided into Thin FYI (30–70 cm, code "7"), Medium FYI (70–120 cm, code "1."), and Thick FYI (>120 cm, code "4."). Based on the state of

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TABLE II
SEA-ICE TYPE DEFINITION AND CODE

Stage of development	Thickness(cm)	Ice-type code
New ice	<10	1
Nilas	<10	2
Young ice	10-30	3
Grey ice	10-15	4
Grey-white ice	15-30	5
First-year ice	30+	6
Thin first-year ice	30-70	7
Medium first-year ice	70-120	1.
Thick first-year ice	>120	4.
Old ice	-	7.

sea-ice movement, sea ice can be divided into fast ice and drift/pack ice. Smooth and rough FY/multiple year sea-ice types are discriminated by the surface roughness. Level and deformed ice types are based on the deformation of ice surface. Small ice cake, ice cake, and floe ice are classified by the size of flat piece of ice [27].

Many studies on sea-ice classification have reported a high accuracy. Yet, the sea-ice types classified were different [21], [28], [29]. For example, Leigh *et al.* [28] focused on sea-ice and water classification. Singha and Ressel [21] classified five sea-ice types in Greenland according to the sea-ice stage of development. Liu *et al.* [29] identified seven sea-ice types in the east coast of Antarctica based on sea-ice states: fast ice, multiyear (MY) pack ice, FY pack ice, deformed FY, brash ice, and open water (OW). Our previous study [19] distinguished four sea-ice types in rough development stages: OW, young ice, FY ice, and old ice. It was found that the sea-ice types at near development stages or with similar characteristics were easily misclassified.

When classifying sea-ice types, operational sea-ice products are considered as the most reliable source of information because in-site measured data of sea ice for large area are not available. In particular, ice charts published by CIS, the National Ice Center in USA, the Norwegian Meteorological Institute (MET), Norway, and the Danish Meteorological Institute (DMI) show regional ice distribution with detailed information of sea-ice concentration, development stage, and so forth. Yet, CIS ice charts are drawn manually by sea-ice analysts with multiple data sources, including high-resolution radiometers, special microwave imagers, SARs, and ship and helicopter observations, along with highly referenced automatic sea-ice classification methods [22].

In this study, we aimed to identify more refined sea-ice types (seven types) according to the sea-ice stage of development: new ice (NI), gray ice (GI), gray-white ice (GWI), thin FY ice, medium FY ice, thick FY ice, and OW (see Table I).

B. SAR Characteristics in Sea Ice

SAR has the advantage of working independently of sunlight conditions and cloud coverage and thus has been widely used for sea-ice classification. The characteristics of sea ice in SAR images are related to several factors: radar band, polarization mode, incidence angle, and other seasonal factors.

The commonly used SAR data come from C-band satellites such as RADARSAT-2 and Sentinel-1/2, X-band satellites such as TerraSAR-X, and L-band satellites such as ALOS. C-, X-, and L-band satellites work at different frequencies, and each of them has different capabilities of penetration depth and sensitivity to surface roughness. Comparative studies have found that C-, X-, and L-band SAR images with the full polarization have similar performance on sea-ice characterization [24], [30]. However, compared to C-band, L-band SAR images could improve sea-ice motion estimates during both the melt and freeze-up time periods [31].

Different scattering mechanisms lead to different polarization modes. Single-polarization SAR provides one channel of intensity data in either horizontal linear transmission and horizontal linear reception (HH), horizontal linear transmission and vertical linear reception (HV), vertical linear transmission and horizontal linear reception (VH), or vertical linear transmission and vertical linear reception (VV) mode. Dual-polarization SAR provides two channels of intensity and phase information, e.g., HH/HV, VV/VH, or HH/VV. Full polarization provides all four channels. Compared with single-polarization and dual- and full-polarimetric SARs can provide more information for sea-ice classification and are more used for multiple-ice-type classification [24], [32], [33]. However, their narrow swath width is a major limitation for monitoring the state of sea ice [34].

Recently, compact polarimetry (CP) of C-band SAR was investigated in major sea-ice type classification [21], [35], [36]. An analysis showed that the features of CP and full polarimetry SAR over the sea surface exhibited similar trends with just subtle differences [3]. However, full and CP data are not always available. Meanwhile, dual-polarization data have been demonstrated to be sufficient to perform the sea-ice classification [37]. Different sea-ice types also show high variations in backscattering features, grayscale, and texture, depending on incidence angle [6], weather conditions, and season [29], [35], [38].

Dual-polarization SAR images (HH+HV) were used in our study, acquired from both melting and freezing seasons.

C. Sea-Ice Classification Methods With SAR Images

Based on the type of SAR features, the sea-ice classification methods can be summarized into two categories: polarimetric feature-based methods and image feature-based methods.

The polarimetric feature-based methods describe the characteristics of sea ice from the perspective of scattering mechanisms or statistical properties. Eigendecomposition is a common way to derive eigenvalues or eigenvectors from the scattering matrix [28], [39], [40] and to derive entropy/ anisotropy/α-angle (H/A/α-angle) parameters from the coherency matrix [2], [41], [42]. Directly using the co- and cross-polarization ratio, the phase difference between the HH- and VV-polarized channels also contributes to the sea-ice

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classification [5], [43]. For the image feature-based methods, texture analysis is most popular via gray-level co-occurrence matrices (GLCMs) [6], [8], [28], wavelet-based features [44], Gabor filter [45], and MRFs [4], [46].

After computing the polarimetric features or image features, a classification method is applied to classify the sea-ice types. The commonly used classifiers include SVM [6], [28], [29], [40], neural networks [23], [24], [47], [48] and maximum likelihood [45]. Besides, image segmentation is frequently used to enable the nearest neighbor pixels to be assigned to the same class with a high probability. The MRF model and the Iterative Region Growing Using Semantics (IRGS) model are often used in operable sea-ice segmentation and classification in recent years [4], [22], [49], [50].

The combined use of different features and classifiers is more effective. Liu *et al.* [29] used TerraSAR-X single-polarization data with radar backscatter coefficients, textures features, and incidence angles to train a seven-ice-type SVM classifier and achieved an average accuracy above 89.71%. Leigh *et al.* [28] proposed the map-guided ice classification that combined the IRGS approach with a modified energy function to accommodate the SVM pixel-based information, and the ice-water discrimination accuracy of 96.42% was achieved. Zhu *et al.* [51] integrated multiple features of polarization ratio and GLCM into the conditional random field (CRF) framework and used a final SVM-CRF model to achieve overall 92% accuracy for five-type sea-ice classification.

At present, there are a few related works on the application of DL methods for sea-ice research. In 2016, Wang *et al.* [16] used a CNN to estimate ice concentration with SAR images. It produced more detailed ice concentration maps than those produced by operational models and more reasonable ice concentration estimations in melting regions and in regions of low ice concentration. In 2017, Wang *et al.* [17] found that CNNs were less sensitive to pixel-level details than multiple-layer perception (MLP) methods, and the results of CNNs produced ice concentration were less noise and closer to Ice Charts. In 2017, Petrou *et al.* [52] first investigated the potential of recurrent neural networks (RNNs) in predicting the ice motions in the next several days. With previously observed satellite image data, their approach could provide satisfactory results for predictions as far as ten steps into the future. Chi and Kim [18] compared MLP and LSTM with the traditional autoregressive (AR) model on predicting the concentration of Arctic sea ice. Their method statistically and visually outperformed traditional AR models. In 2019, Poliyapram *et al.* [53] used a UNet deep CNN structure with dilated convolutional layers to classify water, ice, and land. The method was claimed to be effective in the sunglint affected regions and small water regions.

III. Study Area and Data Preparation

In this section, we describe the details of our data preparation, including the study area location, the data source of SAR images, and the data set construction for model training and testing.

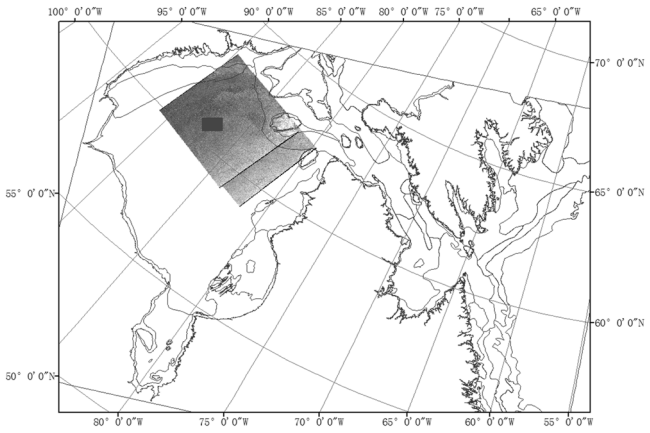


Fig. 2. Study area in Hudson Bay. Most time-sequence data samples came from the area in black square.

A. Study Area

Hudson Bay is chosen as the study area because it is one of the five regions where CIS carries out long-term sea-ice monitoring. Moreover, sea ice usually begins to melt in May and to freeze in October in Hudson Bay, and during the period, different sea-ice types appear alternately. The study area covers the region between 59° 29' 15.29" N–63° 17' 3.22" N and 81°42'36.75"W–90°5'24.54"W.

The study area is observed by Sentinel-1A and Sentinel-1B satellites, which provide day-and-night SAR imagery at C-band. The Sentinel-1 A and B satellites, managed under the European Space Agency (ESA) Copernicus program, have a revisit period of 12 days each, and they together can image the entire Earth every six days. These satellite data are freely available and widely used in Arctic sea-ice monitoring and marine environmental monitoring. Fig. 2 shows two example scenes of SAR images in the study area, taken in two sequential weeks by Sentinel-1A and Sentinel-1B, respectively. It is noted that the two SAR images do not precisely coincide. To maintain the data consistency between the two satellites, we cut off the nonoverlapped areas from data construction in this study.

B. Data Source

The SAR images used in this study are Sentinel level-1 Ground Range Detected (GRD) product, dual-polarized (HH and HV) Extra Wide (EW) swath mode with medium resolution, which have a 400-km swath with the spatial resolution of 20 m $\times$ 40 m. The level-1 GRD products are processed by internal calibration, Doppler centroid estimation, and range and azimuth processing. They consist of focused SAR data that have been detected, multilooked, and projected to ground range using the WGS84 Earth ellipsoid model. More information about level-1 data product can refer to the web https://sentinel.esa.int/web/sentinel/user-guides/sentinel-1-sar/product-types-processing-levels/level-1.

Our experimental Sentinel-1 SAR images were downloaded from the Copernicus Open Access Hub (https://scihub.copernicus.eu/dhus/#/home). The pixel value of these images is in the data type of uint16. A total of 72 Sentinel-1 dual-polarized SAR images were collected

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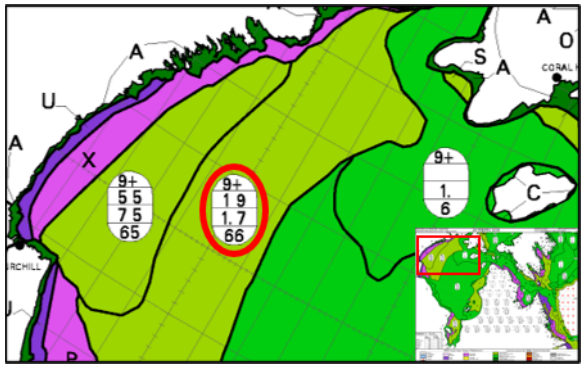


Fig. 3. Weekly Ice Chart in Hudson Bay published by CIS on February 26, 2018. The ice egg marked with a red oval indicates that over 90% of the area is concentrated by sea ice, and the thin FY ice (code 7) with the proportion of 90% is dominated.

from the time periods of October 2017–July 2018 and July 2019–June 2020. These SAR images cover all types of sea ice in Hudson Bay, including OW, NI, GI, GWI, thin FY ice (ThinFI), medium FY ice (MFI), and thick FY ice (ThickFI).

CIS provides daily and weekly regional ice charts that are available at https://iceweb1.cis.ec.gc.ca/Archive/page1.xhtml. According to CIS, the ice charts are drawn on geographical information systems, allowing precise data observation and quick transmission. It is estimated that the geometric accuracy of the ice edge will be within 630 m for an image with 100-m pixel resolution. The ice charts presented in color follow the World Meteorological Organization (WMO) standards [27], and they are coded with the "egg code," providing the sea-ice information of concentration, stage of development (sea-ice type), and floe sizes. Fig. 3 shows a weekly sea-ice chart in Hudson Bay on February 2018. Taking the egg code circled in red as an example, it gives the ice information of the region where the egg locates; 9+ stands for the regional ice concentration, that is to say, over 90% of the region is covered by ice. The notations "1." and "7" indicate that there are two types of sea ice: medium FY ice and thin FY ice (see Table II for sea-ice type code). The digits "1" and "9" indicate that the two types of sea ice account for 10% and 90% of the total amount of ice, respectively. The light green color indicates that the region is dominated by thin FY ice. More information about the ice chart can be found on the Government of Canada website [27].

C. Data Sets Construction

To construct our experimental data set, the original 72 SAR images were preprocessed using the Sentinel Application Platform (SNAP) toolbox (http://step.esa.int/main/download/snap-download/). The preprocessing follows a generic workflow of processing Sentinel-1GRD products [54], including precise orbit acquisition, thermal noise removal, radiometric calibration, and range-Doppler and terrain correction. Incidence angle influence was not discussed in this article, due to the lack of standard normalization algorithm and the complexity of determining normalization parameters for different sea-ice types [55]. After preprocessing, the data set construction is performed as follows: selecting specific regions from these images; mapping them with CIS ice charts in the same geographical coordinate system using ENVI software; labeling the regions according to the ice-type information from the egg code annotation; and finally slicing the regional images into patches with the size of 57×57 pixels to form data samples for training and testing. The patch size was determined based on our previous work [19] (more details in Section V-B). In this study, our labels include seven sea-ice types: NI, GI, GWI, ThinFI, MFI, ThickFI, and OW.

According to our model design (see details in Section IV), we have made two data sets: the ResNet data set for spatial feature learning, and the LSTM data set for temporal feature learning. When training the ResNet network, we wanted to ensure that every sample stood for one single sea-ice type as much as possible to avoid that incorrect sample labels would influence the ResNet model performance. Thus, the samples of ResNet data set came from the regions where the main ice-type concentration was 0.8 and above. For the LSTM network, we needed time-sequence samples to capture the temporal characteristics of the sea-ice type change. Therefore, when building the LSTM data set, we fixed one area of Hudson Bay (e.g., the black rectangle marked in Fig. 2) and clip time-sequential images of the area into patches of 57×57 pixels. The LSTM samples were labeled as the dominant sea-ice type in the corresponding egg code. Table III shows the relevant information of the LSTM data set, which demonstrates the change of sea-ice types over time.

Overall, the ResNet data set includes 100 546 samples, of which the minimum number of sea-ice samples is 5300 (for GWI). The total samples of the LSTM data set are 98 394. All sample data are processed by subtracting the mean and normalization. When using the data sets on a sea-ice classification model, we randomly split them into two training data set and test data sets at a ratio of 7:3.

IV. Methodology

A. Framework of Sea-Ice Classification

Our experimental framework for sea-ice classification is shown in Fig. 4. Given an SAR image of a region at time t, I_t, our goal is to automatically produce a sea-ice distribution map of the region at the time t. The workflow is as follows.

1) *SAR Image Preprocessing:* SAR images covering the same region before the time t are collected to form a time-series image sequence $I_t, I_{t-1}, \ldots, I_{t-s}$. (The required number of SAR images s will be discussed in Section V.) Then, each image is clipped into patches with the size of 57×57 pixels. For a patch P_{ij} in I_t, its position (i, j) and the time of its belonging image (t) should be recorded.
2) *Image Spatial Feature Learning:* For each patch, we use the trained ResNet network to extract its spatial features and output a 512-D feature vector F.
3) *Time Series Feature Learning:* For each time-series patch, we input their spatial features (extracted in step 2) into the LSTM network to further learn their time correlations (e.g., the hidden trend of sea-ice

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TABLE III
DETAILED INFORMATION ABOUT THE LSTM DATA SET CONSTRUCTION

Date	Main ice type, Concentration	Second ice type, Concentration	Date	Main ice type, Concentration	Second ice type, Concentration
20171122	NI, 0.9	——	20200216	MFI, 0.9+	——
20171128	GI, 0.9	——	20200222	MFI, 0.9+	——
20171204	GWI, 0.6	GI, 0.4	20200228	MFI, 0.9+	——
20171210	GWI, 0.9	GI, 0.1	…	…	…
20171216	GWI, 0.9	GI, 0.1	20200323	MFI, 0.8	ThickFI, 0.2
…	…	…	20200329	MFI, 0.8	ThickFI, 0.2
20191206	GWI, 0.9	GI, 0.1	20200404	ThickFI, 0.5	MFI, 0.5
20191212	GWI, 0.9	GI, 0.1	20200410	ThickFI, 0.9	MFI, 0.1
20191218	ThinFI, 0.5	GWI, 0.4	20200416	ThickFI, 0.9	MFI, 0.1
20191224	ThinFI, 0.7	GWI, 0.3	20200422	ThickFI, 0.9+	——
20191230	ThinFI, 0.9+	——	20200428	ThickFI, 0.9+	——
20200105	ThinFI, 0.9+	——	…	…	…
20200111	ThinFI, 0.9-	——	20200603	ThickFI, 0.5	——
20200117	ThinFI, 0.9+	——	20200609	ThickFI, 0.2	——
20200123	ThinFI, 0.9+	——	20200615	OW, 0	——
20200129	ThinFI, 0.8	MFI, 0.2	20200621	OW, 0	——
20200204	ThinFI, 0.8	MFI, 0.2	20200627	OW, 0	——
20200210	ThinFI, 0.5	MFI, 0.5	20200703	OW, 0	——

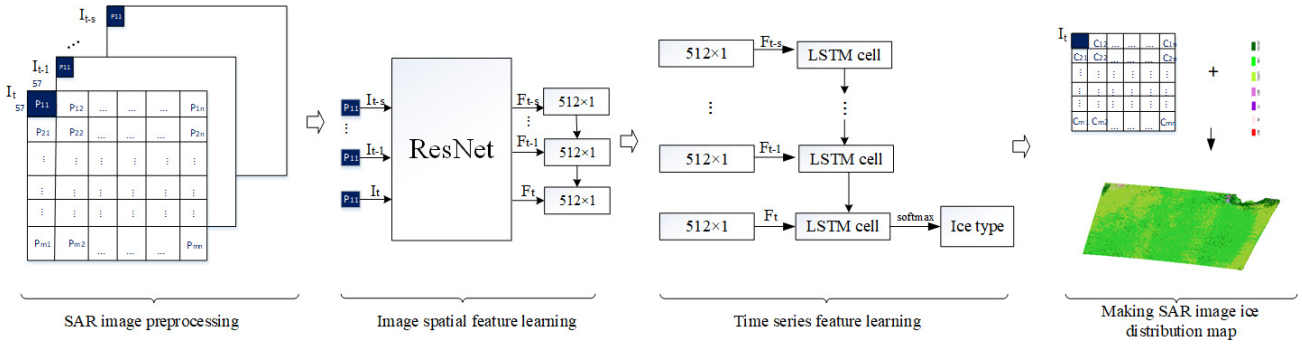


Fig. 4. Framework for sea-ice classification.

development). Then, the sea-ice type of every patch of I_t can be classified by a softmax classifier.

4) *Producing an SAR-Based Distribution Map:* The classification results of patches are restored back to the positions in the original SAR image I_t, and the sea-ice distribution map of the SAR image is generated by mapping the different sea-ice types with the WMO color standard.

In the proposed sea-ice classification framework, the ResNet network needs to be trained independently using the established ResNet data set. To train the LSTM network, the LSTM data set samples are first processed by the trained ResNet network to extract the 1-D feature vectors, which are then used as the input of LSTM units.

B. Residual Neural Network

A residual convolutional network is one of the classic CNN models in DL, mainly applied for solving the degradation problem of deep networks [10]. In this article, we used an 18-layer ResNet, whose structure and configuration follow those in [25], as listed in Table IV.

Residual block and identity mapping by shortcuts are the main characteristics of the ResNet network. The working principle is shown in Fig. 5. Let x be the input image and $H(x)$ be an underlying mapping to be fit by several convolution layers. The residual network defines a residual function $F(x) := H(x) - x$. Then, the original mapping becomes $F(x) + x$. It would be easier to push the residual to zero rather than fitting an identity mapping by a stack of nonlinear layers. One residual block is defined in the following equation:

$$y = F(x, W_i) + W_s x \tag{1}$$

where y is the output vector of the layers. The function $F(x, W_i)$ represents the residual mapping to be learned; W_s is

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TABLE IV
STRUCTURE AND CONFIGURATION OF THE RESNET MODEL USED IN THE PRESENT STUDY

operate	kernel size	stride	output size
input	——		57×57×2
conv1	7×7	2	29×29×64
	BN, Relu		
pool1 (Max)	3×3	2	15×15×64
conv2_x	3×3	1	15×15×64
	BN, Relu		
	3×3	1	
	BN, Relu		
	3×3	1	
	BN, Relu		
	3×3	1	
conv3_x	BN, Relu		8×8×128
	3×3	2	
	BN, Relu		
	3×3	1	
	BN, Relu		
	3×3	1	
	BN, Relu		
	3×3	1	
conv4_x	BN, Relu		4×4×256
	3×3	2	
	BN, Relu		
	3×3	1	
	BN, Relu		
	3×3	1	
	BN, Relu		
	3×3	1	
conv5_x	BN, Relu		2×2×512
	3×3	2	
	BN, Relu		
	3×3	1	
	BN, Relu		
	3×3	1	
	BN, Relu		
	3×3	1	
pool2 (Average)	2×2	1	1×1×512
dense	softmax		7

the linear transformation when the output dimension does not match with the input.

In this article, the 18-layer ResNet has eight ResNet blocks in four convolution layers. Batch normalization (BN) and a ReLU function [56] are used in each layer [25], [57].

The loss function is the cross-entropy cost function, as defined in the following equation:

$$\text{loss} = -\frac{1}{n}\sum_{n} y \ln a + (1-y)\ln(1-a) \tag{2}$$

where a is the predicted output, y is the expected output, and n is the total number of samples.

C. LSTM Network

LSTM [26] is an RNN, which is capable of learning long-term dependencies. It is widely used in machine translation, affective analysis, speech recognition, and other neighborhood fields. The core of LSTM is the cell state. Compared with the traditional RNNs, LSTM uses a gating mechanism to control the selectivity of information (i.e., the connection with the previous state). A memory unit structure of LSTM is shown in Fig. 6, which contains three gating units, namely, forgetting gate, input gate, and output gate.

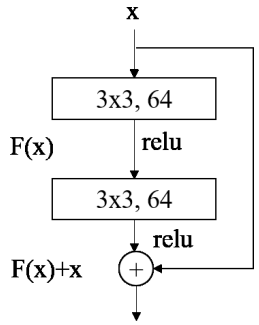


Fig. 5. Residual block.

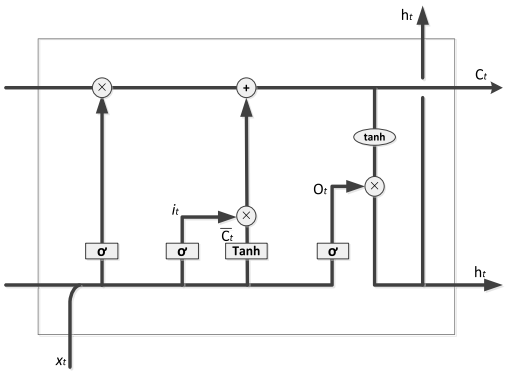


Fig. 6. Example of LSTM cell.

The forgetting gate is used to control what information is discarded from the previous state. The input gate is used to determine what new information the cell state will store. The output gate is used to determine the output value. The three gating units for LSTM are described in the following equation:

$$\begin{aligned} f_t &= \sigma(W_f g[h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i g[h_{t-1}, x_t] + b_i) \\ \bar{C}_t &= \tanh(W_c g[h_{t-1}, x_t] + b_c) \\ C_t &= f_t \cdot C_{t-1} + i_t \cdot \bar{C}_t \\ O_t &= \sigma(W_o[h_{t-1}, x_t] + b_o) \\ h_t &= O_t \cdot \tanh(C_t) \end{aligned} \tag{3}$$

where W and b represent, respectively, the weight and the bias of the corresponding gating, σ is the sigmoid function, f_t is the forgetting gate control signal, i_t is the input gate control signal, $\hat{C}_t$ is the new cell information, C_t is the current cell information, o_t is the output gate control signal, and h_t is the current cell output value.

For different tasks, LSTM has different structures, such as one-to-one, one-to-many, many-to-one, and many-to-many. In the ice-classification task, we use a many-to-one LSTM structure with one hidden layer, as shown in Fig. 7. With the output features of the LSTM, we used the softmax classifier to get the classification result of the input F_t. The units of the

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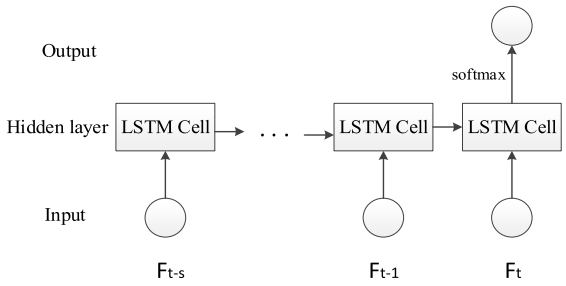


Fig. 7. LSTM's architecture.

LSTM cell are 256, and a common stochastic optimizer Adam is used. The loss function used in the LSTM network is also cross-entropy, the same as that in the ResNet.

V. Experiments and Analysis

We conducted a set of experiments to study the influencing factors of the sea-ice classification performance and apply our model in real sea-ice interpretation.

Our experiments were run on a computer with the following specifications: Windows 10 operating system, Intel Xeon E5-1620 3.5GHz CPU, 64-GB Memory, and NVIDIA Quadro K620 GPU. Keras framework (TensorFlow's high-level API, https://keras.io/) and Python programming language were used to build and train DL networks. Under such configuration, it took 6–7 h to train our proposed model. With the trained model, it took 15–20 min to process an SAR scene, in which 12–15 min for cutting the whole scene into patches, 3–5 min for predicting the sea-ice type, and 1 min for stitching and generating a complete sea-ice distribution map.

The classification performance was assessed with the overall accuracy and Kappa coefficient. Accuracy is defined in the following equation:

$$\text{accuracy} = \frac{m}{n} \times 100\% \tag{4}$$

where m is the number of correctly classified samples and n is the total number of samples.

The Kappa coefficient is a statistic often used to measure the consistency of a multiclassification model. It can be calculated using the following equation:

$$k = \frac{p_o - p_e}{1 - p_e}$$
$$p_o = \frac{m}{n}, \quad p_e = \frac{\sum_{i=1}^{7} a_i b_i}{n^2} \tag{5}$$

where k stands for the Kappa coefficient, p_o is observed agreement among the classified samples, p_e is the hypothetical probability of the chance agreement, and a_i and b_i are the number of true samples and the number of predicted samples for the ith sea-ice type, respectively.

A. Influence of Polarization Mode of the SAR Data Set (HH vs HV vs HH + HV)

Some literature shows that different SAR image polarizations have different recognition abilities for certain sea-ice types [24], [32], [33]. However, our previous study of CNN-based sea-ice classification only used single-polarization SAR images [19]. Therefore, we first investigated the influence of different polarization modes of SAR image on classification performance with ResNet. Single-polarization mode HH or HV and dual-polarization mode HH + HV were used in the test. An HH + HV data sample has the HH and HV in two channels and is processed as a whole.

TABLE V
Classification Performance in Different Polarization Modes (%)

Train data	HH	HV	HH+HV
Accuracy	73.98	71.02	86.75

The experimental results are shown in Table V. Dual-polarization mode HH+HV achieves much better classification performance than single polarization, gaining over 14% in accuracy. It indicates that the combination of multipolarization data can significantly improve the recognition ability of sea ice. The difference between HH and HV is not significant. The seven-type sea-ice classification accuracy with HH mode is 73.98%, which is much lower than the accuracy achieved in our previous four-type sea-ice classification [19]. This indicates that finer sea-ice type classification requires much more information from SAR images.

Next, we investigated how the different polarization modes contribute to the identification of different sea-ice types. Fig. 8 shows the classification results of the seven sea-ice types with different polarization modes. Overall, GWI and ThinFI achieved the lowest accuracies. This could be explained by the close development stage of these two sea-ice types: 15–30 cm for GWI and 30–70 cm for ThinFI.

HV polarization was better than HH only for identifying OW. For NI, GI, and GWI, the classification accuracies using the HH mode are significantly higher. These findings agree with those from previous studies [40], [58], which suggested that HH provided a good contrast between NI and GI due to its strong sensitivity to surface scattering differences [40], whereas HV was useful in distinguishing rough thin ice from OW because of its being less sensitive to wind speed [58]. For classifying MFI and ThickFI, our study does not show much difference between HH and HV modes. Although HV individually did not benefit in identifying most of the sea-ice types, combining HV with HH brought a remarkable increase in classification accuracy, which should be attributed to the complementary of each other's advantages. In particular, the accuracy of the GWI type, which was only 47% and 58% for HV and HH, respectively, reached 80% for HH + HV mode. This implies that the interfeatures between HH and HV, learned by the ResNet, provide important mutual information that is beneficial to identify the most difficultly-to-recognize sea-ice types.

The discussions defined next are all based on dual-polarization data.

B. Hyperparameters Design of the Classification Model

The hyperparameters of a DL network are often studied in order to rain the network, striving for optimal results. In our

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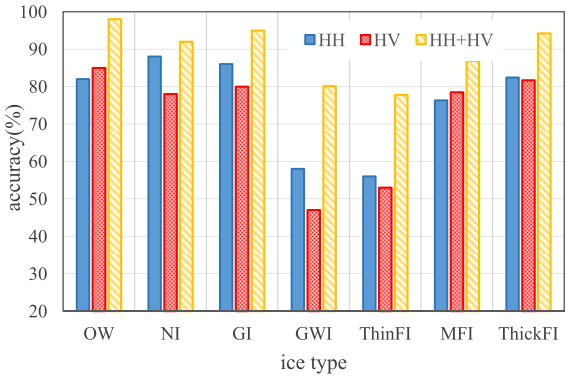


Fig. 8. Classification accuracy for seven-ice type with different polarization modes.

TABLE VI
ACCURACY OF DIFFERENT TIME STEPS ON LSTM

Time step	1	2	3	4	5	6
Acc (%)	60.06	77.04	85.40	90.44	93.45	95.69

previous work on a ResNet-based sea-ice classification model with SAR images, we have determined that the optimal patch size was 57×57 pixels and 4000 samples per class were sufficient for training the model [19]. Thus, in this work, we simply adopted the 57×57 pixels as the input patch size and included a minimum of 5000 samples for each sea-ice type. The network parameters, such as the number of layers and the size of filters, were the same as those in [25], given that the network structure was good enough to perform general classification tasks. Also, we found experimentally that there was no gain from increasing or decreasing the network layers.

The LSTM network can learn the time dependencies between different sea-ice types. We tested the effect of the LSTM time steps on the performance of sea-ice classification. According to our experimental flow design, the entire training process of an n time-step LSTM is shown as follows: 1) group every n samples in the LSTM data set to form sample sequences; 2) randomly select 70% of sample sequences as the training data and the rest as the testing data; 3) for each of the sample sequences, extract $512 \times n$-dimensional feature vectors through the trained ResNet network; and 4) input the feature vectors of the training data to train the LSTM network and then test the classification accuracy of the trained LSTM with the feature vectors of the testing data.

The classification experiments were carried out with time steps varying from 1 to 6 (see Table VI). We can see that a higher accuracy can be achieved with the increase of time steps of the LSTM network. When the time step is set to 6, the classification accuracy of the test set reaches 95.69%.

C. Comparison of Sea-Ice Classification Models (ResNet vs ResNet + LSTM)

It is much harder to classify the more similar sea-ice types, and this works for two reasons. On the one hand, different sea-ice types defined by the thickness of sea ice present very similar visual features in SAR image, and they are difficult to recognize by image segmentation methods. On the other hand, because of the different resolutions of SAR images, one pixel of SAR image may represent tens of squared kilometers of sea area. Therefore, when we cut the image into sample patches, it is easy to mix different sea-ice types into one sample, especially at the boundary between different types of sea ice. The design of our sea-ice classification framework using ResNet and LSTM networks aimed to solve these problems.

TABLE VII
CLASSIFICATION CONFUSION MATRIX OF RESNET

	OW	NI	GI	GWI	ThinFI	MFI	ThickFI
OW	1.00	0.00	0.01	0.00	0.00	0.00	0.00
NI	0.00	0.84	0.001	0.00	0.04	0.04	0.07
GI	0.00	0.02	0.96	0.01	0.01	0.00	0.00
GWI	0.00	0.02	0.01	0.76	0.18	0.00	0.00
ThinFI	0.00	0.01	0.00	0.01	0.70	0.21	0.07
MFI	0.00	0.00	0.00	0.00	0.13	0.65	0.22
ThickFI	0.00	0.00	0.00	0.00	0.05	0.17	0.78

TABLE VIII
CLASSIFICATION CONFUSION MATRIX OF ResNet + LSTM

	OW	NI	GI	GWI	ThinFI	MFI	ThickFI
OW	0.997	0.00	0.00	0.00	0.00	0.002	0.00
NI	0.00	1.00	0.00	0.00	0.00	0.00	0.00
GI	0.00	0.00	1.00	0.00	0.00	0.00	0.00
GWI	0.00	0.00	0.00	0.99	0.001	0.00	0.00
ThinFi	0.00	0.00	0.00	0.00	0.94	0.05	0.01
MFI	0.00	0.00	0.00	0.00	0.02	0.94	0.03
ThickFI	0.00	0.00	0.00	0.00	0.00	0.05	0.95

In the ResNet training data set, the samples are all from the regions where the main sea ice has a concentration ratio of 0.9 and above, which keeps the purity of sea-ice type. With this data set, the ResNet network can reach 86.75% classification accuracy. However, in the LSTM data set, the sea-ice regional concentration varies from 0 to 0.95, which is closer to the real SAR classification samples. Thus, with the LSTM data set, the ResNet's classification accuracy drops to 79.40%, with the Kappa coefficient of 0.72.

Table VII shows the confusion matrix of sea-ice classification results, using the ResNet network with the LSTM data set. It can be seen that the misclassified samples are mostly in the sea-ice types at adjacent development stages, e.g., the ratio of misclassified MFI samples is 0.13 for ThinFI and 0.22 for ThickFI. On the contrary, using ResNet + LSTM, various ice types have been classified with high accuracy and high Kappa coefficient (0.97), as shown in Table VIII.

By the comparison between Tables VII and VIII, it is apparent that the LSTM network improves the classification accuracy of easily confused sea-ice types, such as ThinFI, MFI, and ThickFI. This is owed to our approach that utilizes the time dependence of sea-ice types. Taking ThinFI as an example, Fig. 9 shows that the ratio of ThinFI being misclassified into other sea-ice types decreases as the LSTM time steps

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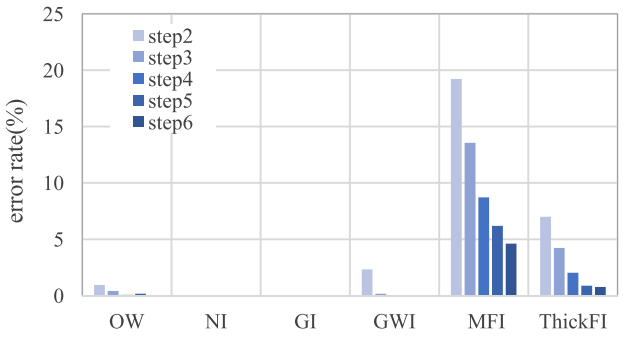


Fig. 9. ThinFI's misclassification rates with the different time steps.

TABLE IX
CLASSIFICATION RESULT OF DIFFERENT METHODS

	MLP	SVM	ResNet	ResNet+LSTM
OW	100.00	100.00	100.00	99.76
NI	42.37	35.86	84.15	100.00
GI	63.54	75.42	96.33	100.00
GWI	62.83	70.93	76.23	99.90
ThinFI	46.10	48.75	70.36	94.52
MFI	72.25	77.33	64.60	94.03
ThickFI	33.68	41.27	78.08	95.13
Accuracy	**50.17**	**60.30**	**79.40**	**95.69**
Kappa	**0.45**	**0.50**	**0.72**	**0.94**

increase. In our experiments, learning from six-week changes of ice types (six time steps), our method can more accurately determine the current ThinFI stage.

Overly high accuracy for some ice types (e.g., OW and GI in Table VII) was somewhat to be expected, given the fact that the model was trained and tested on well-curated data sets. Moreover, some ice types are indeed easy to distinguish from others. This was also observed by other studies. In particular, OW can be unambiguously discriminated (>99% accuracy) from the other sea-ice types [40].

D. Comparison With the MLP and SVM Methods

The well-known SVM has been typically used in sea-ice classification with SAR images [58]. Satisfactory classification results have also been achieved with MLP with the features [40], [48]. Thus, we compared our proposed method with these two common algorithms of MLP and SVM.

In the experiments, SVM used radial basis function (RBF) kernel, and its parameter C was set to 1 and gamma was automatically learned. MLP had three hidden layers, 512 neurons in each hidden layer. They were trained with a learning rate of 0.001.

From Table IX, we observe that the accuracies of SVM and MLP reached about 50%–60% and Kappa values were about 0.5. Specifically, both SVM and MLP performed well in ice-water classification, gaining 100% accuracy for OW. For MFI, they were better than ResNet, obtaining a high classification accuracy of above 70%. SVM was also good for GI and GWI. Based on the overall performance, SVM and MLP are still inferior to ResNet when classifying finer sea-ice types of NI, ThinFI, and ThickFI, and they apparently underperform our proposed method based on ResNet+LSTM, which achieved an accuracy of 95.69% and a Kappa value of 0.94.

The training and evaluation of SVM and MLP used the same data set as that of ResNet. SVM is kernel learning method, and MLP is a shallow neural network. They can only learn low-level features of the SAR images for classification. However, the multiple sea-ice types, especially the sea-ice types in the adjacent development stages, have very similar texture and gray level features. This makes the classification methods based on low-level features much difficult. A DL method has the advantage of learning high-level features and thus can achieve a better classification accuracy. LSTM is particularly suitable for capturing the characteristics of time dimension by taking the short- and long-term dependencies into account.

E. Sea-Ice Distribution Map

Previous experiments were conducted with a well-prepared data set. For example, data samples for ResNet were only from the regions having one dominant sea-ice type. The data samples for LSTM were from small regions. To verify that the model is not overfitting the given data set and is applicable in real-world applications, we employed the proposed method for the generation of sea-ice distribution maps. The sea-ice distribution maps are the most intuitive way to understand the sea-ice situation and important for ship navigation. In this experiment, we produced SAR-based sea-ice distribution maps from February 4, 2020 to February 22, 2020 (about four weeks) using both the ResNet and the ResNet+LSTM models. The original SAR images for the experiment were at the resolution of 15 171 × 14 806 pixels and were preprocessed [54] before the usage.

Fig. 10(a) shows the HH- and HV-polarized SAR images from Sentinel-1 satellites. Although the HV polarized SAR images have undertaken noise correction, there is still an obvious subswath effect. This is because cross-polarization channels are more significantly disturbed by thermal noise, for their depolarized power that is naturally weaker than initially polarized power [59]. The roughly corresponding areas of these SAR images are marked in red boxes in the CIS ice chart, as shown in Fig. 10(b). Fig. 10(c) shows the sea-ice distribution maps based on ResNet classification results, and Fig. 10(d) shows the maps based on ResNet+LSTM classification results. It should be noted that the maps of Fig. 10(c) and (d) cover a bit smaller area than the SAR images of Fig. 10(a). The six time-step LSTM networks require six images from the same region, which are alternatively from the Sentinel-1 A and Sentinel-1 B satellites at an interval of six days. As the SAR images from the satellites A and B have dislocation in terms of latitude and longitude, we selected the largest overlapping part of the six images to make the sea-ice distribution map. Besides, since the sea-ice types in this article do not include coastal ice-free land, we removed the small coastal part at the top-right corner from the output maps of Fig. 10(c) and (d).

Because there is no ground-truth map that can provide pixel-level identification of sea-ice types, the sea-ice classification results can only be discussed based on the similarity with the ice charts. However, it should be noted that the ice

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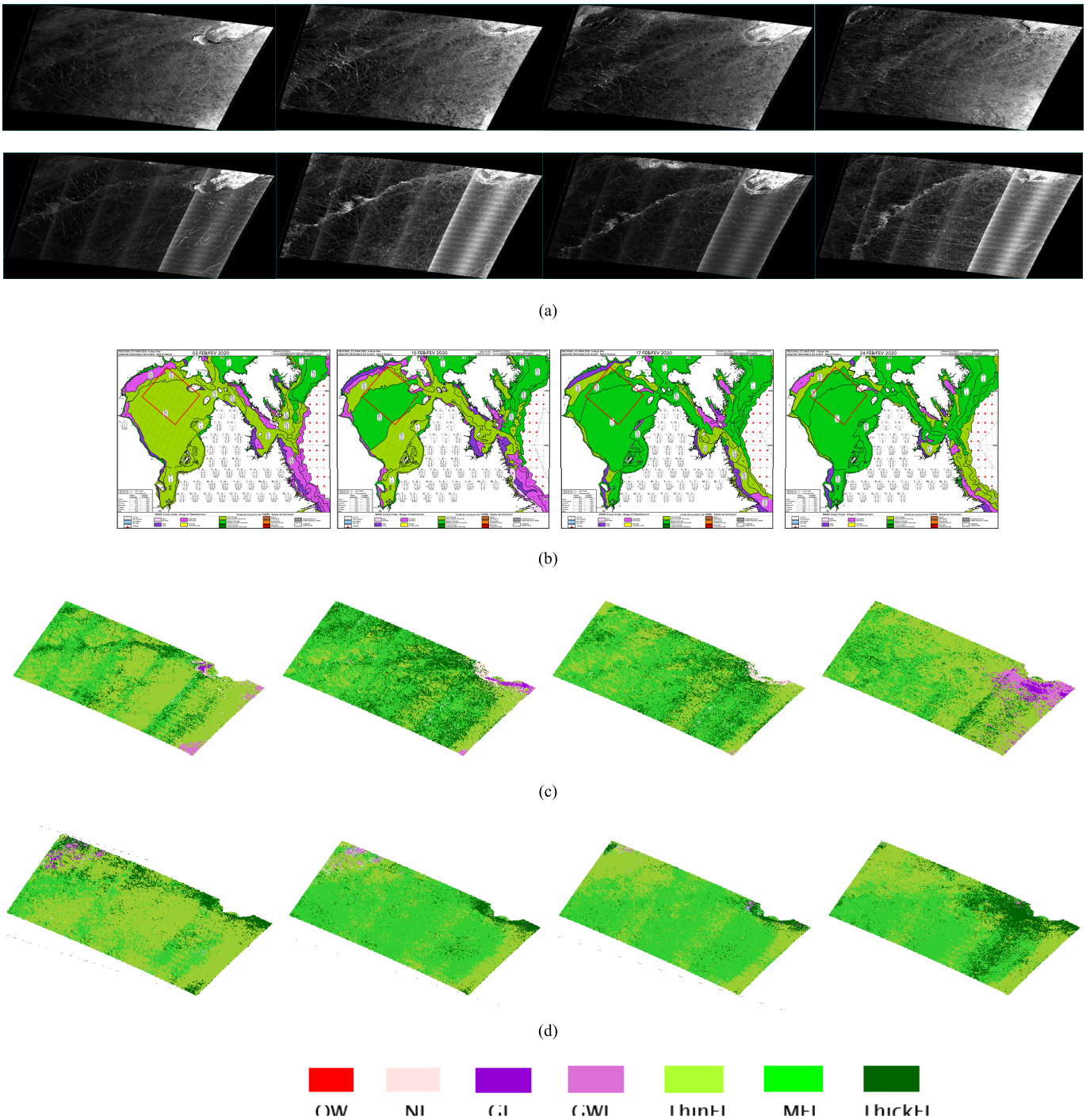


Fig. 10. Ice distribution maps based on the preprocessed SAR images. The acquired time of original SAR images is February 4, 2020, February 2, 2020, February 16, 2020, and February 22, 2020 from left to right. (a) SAR images after preprocessing: the first row shows HH polarized images and the second row shows the HV polarized images. (b) Four weekly ice charts by CIS: the red rectagulars stand for the covering range of the original SAR images. (c) Classification results using the ResNet network. (d) Classification results using the ResNet + LSTM network.

charts in Fig. 10(b) are not on the exact same dates as the SAR images. They were weekly released on February 3, February 10, February 17, and February 24, 2020.

The strong subswath effect of HV polarized SAR images, shown in Fig. 10(b), affected the classification results. This can be observed from the changes of sea-ice types (between ThinFI and MFI) at the interswath regions in Fig. 10(c) and (d). However, the results using ResNet + LSTM method are less affected than those using the ResNet method. According to the CIS ice charts, the overall concentration of sea ice in the study area was 9+, which means that there was almost no OW. Both ResNet and ResNet + LSTM networks did not produce OW in Fig. 10(c) and (d). These results confirm the high classification accuracy of OW in Table IX.

Based on the ice charts, the dominant sea-ice type on February 4, 2020 was ThinFI (shown in light green color), and then, MFI (shown in green color) increased from February 10, 2020 to February 16, 2020. The maps produced by ResNet +

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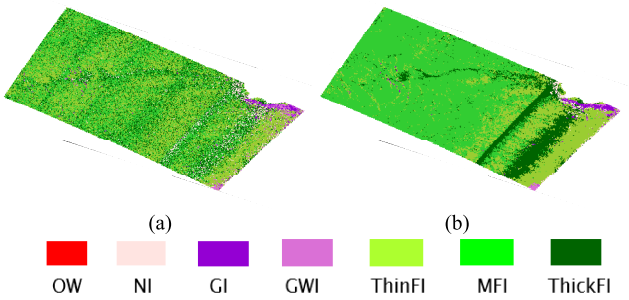


Fig. 11. Ice distribution map based on (a) MLP and (b) SVM, the acquired time of SAR image is February 16, 2020.

LSTM are generally consistent with the ice charts, except ThickFI (shown in dark green) that appears in the fourth week map. This might be explained by the fact that LSTM is a time recursive neural network, and it can predict the stage development of sea ice based on the long- and short-term dependencies of time series. It is highly possible that MFI will be developed into ThickFI soon. At the top-left corner of the first two maps in Fig. 10(d), we can also observe a little GWI. GWI is actually reported in the ice egg of the corresponding ice charts, taking up 20%–30% of the small corner. The main errors of the maps generated by ResNet + LSTM locate around the edge of the coastal land. We cannot remove the coastal region completely, and thus, the result is affected by the brightness of coastal regions on SAR images. In future work, involving the ice-free land as a class can solve this problem.

ResNet is less accurate than ResNet + LSTM. The second ice distribution map on February 10, 2020 in Fig. 10(c) shows too much ThickFI, which does not accord with the trend of sea-ice development. It can be explained by the high misclassification rate of MFI in Table VII, where MFI is mostly misclassified into ThickFI. Although ResNet + LSTM is based on the spatial features learned by ResNet, it is able to correct the misclassification problem of ResNet and produces reasonable results. It indicates that the combination of spatial and temporal features using the ResNet + LSTM network can effectively improve the accuracy of sea-ice classification.

Fig. 11(a) and (b) shows the ice distribution maps using MLP and SVM, respectively, with the SAR image acquired on February 16, 2020 [the third figure of Fig. 10(a)]. Due to high accuracy of classifying OW (100%) and MFI (above 72%) as listed in Table IX, both MLP and SVM can successfully recognize the sea-ice areas and determine the dominant sea-ice MFI. However, the result of SVM is clearly affected by the subswath effect of HV polarized SAR images, whereas the result of MLP is sparked by the sea-ice types of ThinFI and ThickFI and is lack of spatial consistency. Comparing Fig. 11 with Fig. 10(c), we cannot get the conclusion that ResNet is better than MLP and SVM, although ResNet reached a higher accuracy on certain sea-ice types (e.g., NI, GI, and GWI in Table IX). From the comparison between Figs. 10(d) and 11, the DL model based on spatial and temporal features is more suitable than MLP and SVM in the classification of various sea-ice types.

VI. Conclusion

In this article, we presented a DL-based automatic sea-ice classification with SAR images. The proposed method combines ResNet and LSTM networks to learn spatial and temporal features of sea ice, directly from the time-series SAR images. It is able to classify seven ice types in different stages of development, including OW, NI, GI, GWI, thin FY ice, medium FY ice, and thick FY ice, achieving an overall accuracy of 95.7% with our data set. With the trained DL models, we could deploy this DL-based sea-ice classification framework for ice interpretation tasks and provide ice distribution maps consecutively and automatically, which could considerably reduce costs and human bias.

Comparing our results with those by SVM and MLP, we found a relatively low classification accuracy of SVM and MLP. This may be explained by the following reason: sea-ice types, especially located in the adjacent development stages, have very similar texture and gray-level features on SAR images, which makes the classification methods based on low-level spatial features ineffective. Our proposed DL method has the advantage of learning high-level spatial and temporal features and thus can achieve a better classification accuracy.

In this article, we adopted ResNet and LSTM to learn the spatial and temporal features of sea ice, respectively. There are many different DL architectures that can be used, but it is impossible to try all of them to find the best one. Using ResNet as the spatial feature extractor was based on our previous study [19], in which ResNet performed better than a typical CNNs. LSTM has been chosen for learning from time-sequential data due to its capability of remembering long-term dependencies of data. GRU [60] simplifies the LSTM architecture and can be trained faster; however, LSTM outperforms GRU in the tasks of modeling long-distance relations with sufficient training data. In this study, six-week development of sea ice needs to be learned, and therefore, we chose LSTM as the model for temporal feature extraction.

While focusing on a learning-based method for automatic ice classification with SAR images, we found that SAR image preprocessing on calibration, speckle, and thermal noise reduction, range-Doppler and terrain correction could influence the classification results. With the SAR image preprocessing, the subswath effect of cross-polarization channel (e.g., HV) could be reduced, but not largely eliminated. Other more effective preprocessing of incidence angle normalization and thermal noise reduction should be executed.

There are still some limitations related to data sources, data set construction, and model parameters. In this study, coastal land has not been taken into consideration, but it often appears in sea-ice areas. Therefore, the future study needs to distinguish land from sea-ice types. As our classification method was performed at the pixel-patch level, it is inevitable that mixed sea-ice types are included in some pixel patches for the actual SAR image classification. A potential solution that we will explore is to involve mixed-type samples into the data set construction and use multiple labels as the ground truth. Due to the six-day revisit period of the Sentinel satellites, our method was developed for producing weekly sea-ice

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distribution maps. During freeze-up and melting periods, sea ice often changes quickly, and more frequent sea-ice classification will be needed.

In this study, the parameters of the DL model were mainly determined based on earlier studies. These may not be the optimal parameters, potentially affecting the model performance. However, our experimental results indicated that these parameters were appropriate for the convergence of model training and a high prediction accuracy.

Our method was established with Sentinel SAR images at C-band. We will examine the applicability of the method with other SAR images from different satellites and bands, e.g., RADARSAT.

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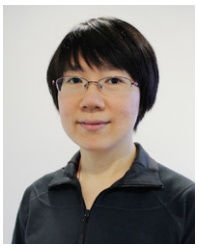
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